

HAT-P-36b

R-0875

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240303 · created 2026-07-28T23:56:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460372.8225 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.022
Transit depth (Rp/Rs)^2	1.97 +/- 0.62 [%]
Orbital Inclination (inc)	81.2 +/- 2.62
Airmass coefficient 1 (a1)	1.189 +/- 0.0097
Airmass coefficient 2 (a2)	-0.138 +/- 0.033
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	None
Optimal Aperture	3.66
Optimal Annulus	4.13
Transit Duration (day)	0.056 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.022 vs 0.126 (deviation 0.64 σ)
Duration fit vs published	0.056 d vs 0.0925 d (deviation 3.32 σ)
Mid-time O-C	-13.8 min
Depth SNR	6.4
Residual scatter	0.81 %

Light curve

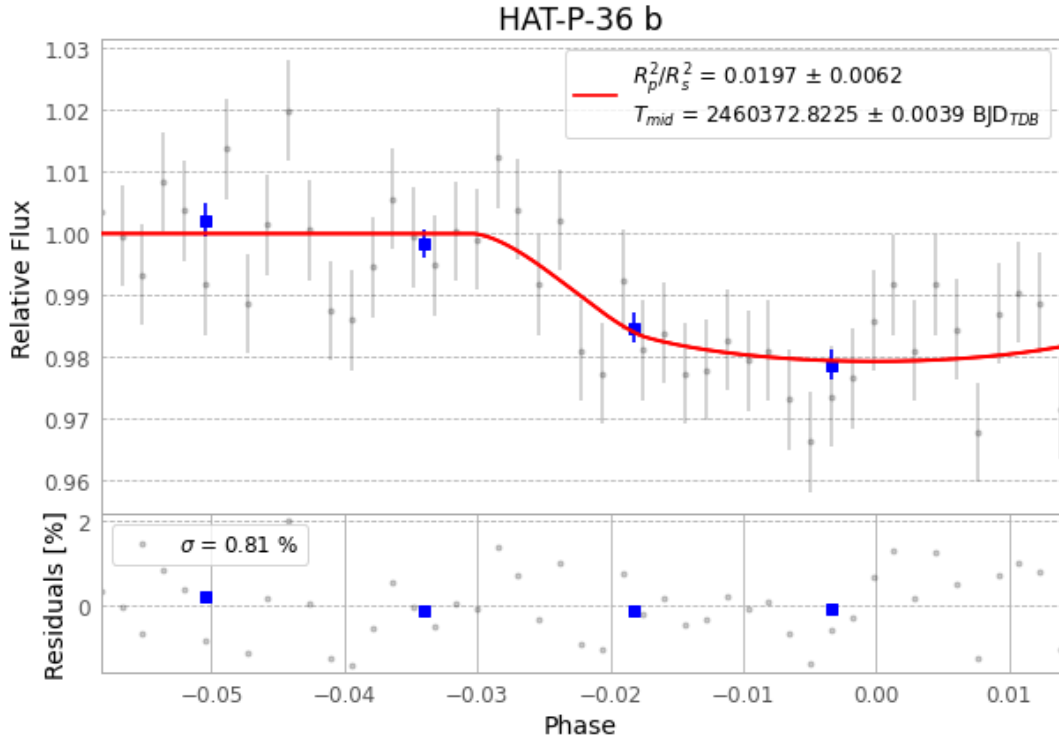


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-03-02.png (SHA-256 438d3c3b80ecbef1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0c4949db7fae1a97...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

47 FITS frames (31 MB), HATP-36240303054515.FITS → HATP-36240303080315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240303.log (c1fc5a2f1824...), FinalLightCurve_HAT-P-36 b_2024-03-02.png (438d3c3b80ec...), FinalLightCurve_HAT-P-36 b_2024-03-02.csv (4b064c56c989...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 23:56Z.